increasingly common in the chemical industry. The principle of all such equipment is illustrated by Fig. A.1(a); it contains a mixing section for efficient transfer of materials between the phases and a settling section for efficient phase separation. The inputs and outputs provided for connecting stages in the counter current extraction scheme (Fig. A.2). *Mixer-settlers* (Fig. A.1(a)) provide good mixing and reasonably good phase separation performance but rather large hold-ups.

For reprocessing the equipment must be highly reliable, have high stage efficiency, short contact times, small liquid inventory (hold-up), be easy to decontaminate and to service, and not least be safe against criticality. High reliability usually means simple design and few (if any) moving parts. The *packed columns* meet this requirement. They are simply long columns (often 10–20 m with a diameter of 0.3–3 m) filled with small pieces of material obstructing a straight flow through the column, which is by gravity (Fig. A.1(b)). However, high stage efficiency requires mechanical agitation of the two phases and clean phase separation, which cannot be met by packed columns.

In *pulsed columns* (Fig. A.1(c)) the mechanical agitation provides good mixing but poor phase separation. Each plate is perforated (a sieve). The organic and aqueous phases separate between the plates ('settling chambers'). In the down movement (<¼ of the interplate distance) the aqueous phase is forced through the sieves, forming droplets, which by gravity fall through the lighter organic phase and coalesce when reaching the interface boundary. In the upward stroke, organic droplets form and rise through the aqueous phase until they meet the organic phase boundary.

The phase separation in a mixer-settler battery or in a column is usually not better than ~99%, i.e. each outgoing phase contains some percent entrained droplets of the other phase. This separation efficiency can be improved to almost 100% by using *centrifugal extractors* (Fig. A.1(d)). Centrifugal extractors affect good mixing, good phase separation, and have very small hold-ups. The organic-aqueous phase contact time in centrifugal extractors can be made much shorter than in mixer-settlers or columns. The small hold-up volume and short residence time cooperate to reduce radiation decomposition.

Packed columns were used in the first Windscale plant (Sellafield, UK). Pulsed columns were used at Hanford (USA), in the old Eurochemic plant at Mol (Belgium), and are currently in use in the newer La Hague and THORP plants. Mixer-settlers have been used at Savannah River (USA), in the Magnox plant at Sellafield (UK), and at La Hague (France). Centrifugal extractors have been installed at Savannah River and at La Hague.

A.5. EXERCISES

A.1. In a solvent system the distribution ratio, D_U, is 2 for uranium and D_{Cs} is 0.003 for caesium. If 99.5% U is to be extracted in a repeated batch fashion (Eqn (A.5)), (a) how much Cs is coextracted? If instead a counter current process is used with

10 extraction and 2 wash stages, what percentage of (b) uranium and (c) caesium is extracted? In an extraction equipment D_{Cs} 0.003 cannot be maintained, because droplets are carried over between stages; the practical value will be D'_{Cs} 0.02. (d) How much caesium is extracted in this latter case with the counter current equipment? Assume equal phase volumes.

A.2. After scrubbing the solvent (1M TBP in kerosene) with a carbonate solution it should be reacidified to equilibrium with 6 M nitric acid before reuse in reprocessing operations. Assume that the partial molar volume of nitric acid can be neglected for both the organic phase and the aqueous phase. Thus, in this example extraction of nitric acid will not increase the volume of the organic phase nor reduce the volume of the aqueous phase. Extraction of nitric acid is assumed to only occur as the complex $TBP \cdot HNO_3$. For simplicity we will assume that the Eqn $D_{HNO_3} = k_{ex} \times [HNO_3]_{aq} \times [TBP]_{free,org}$, with $k_{ex} = 0.1$ can be used to calculate the D-value for nitric acid. The reacidification will be performed in a simple counter current mixer-settler battery. (a) How many stages and (b) what flow rate ratio (org/aq) would be needed in order to produce the required nitric acid concentration in the organic phase using the smallest possible feed of 6.4 M nitric acid? (c) What would be the concentration of the spent nitric acid?

A.6. LITERATURE

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